Crypto Thin Losers

Strategy report · Research desk · 26 October 2026

Status: passed, ambiguous test zone · *full-sample t 2.19*

Abstract

Crypto Thin Losers buys the least liquid coins among recent losers on Binance USDT perpetuals and sells the most liquid ones, with a BTC hedge. Over January 2021 to September 2026 it earns a net Sharpe ratio of 0.92 (90 % bootstrap range 0.25 to 1.52), with no exposure to BTC. On the sealed test, January 2025 to September 2026, the net Sharpe falls to 0.38 (range −0.88 to 1.51). **Three findings weigh against it: in the test the profit came from the short leg, not from the recovery the hypothesis predicts; the gains sit in very few days and names; and the book was under water for 884 days.** It stays in the library on paper only.

1 Hypothesis and method

When a thin coin falls, its sellers meet few buyers and push the price below value; whoever absorbs the flow is paid by the recovery. The pattern comes from Begušić and Kostanjčar (2019), measured in-sample on spot coins from 2015 to 2019, before costs.

The universe is the Binance USDT perpetuals listed for at least 180 days, point in time, with delisted names kept to their last price. Each week the losers are sorted on illiquidity; the book is long the thin losers, short the liquid losers, and hedged on BTC. Costs are charged by tier (20 bp on the thin leg, 5 bp on the liquid leg and the hedge), with funding and liquidations. The train runs from 2021 to 2023, the validation over 2024, and the test from January 2025; the test was opened once, on 24 October 2026.

2 Performance

The full sample includes the train, on which the strategy was designed; the test is the honest estimate. Over the test the strategy returned 9.0 % a year for 23.8 % volatility, and on 21 months a Sharpe ratio of 0.38 can be told apart neither from zero nor from the 0.92 of the full sample. The strategy made 38 %, 37 % and 32 % in 2021, 2022 and 2023, then about 8 % a year. The maximum drawdown, −38.4 %, runs from April 2024 to April 2025, and the peak was only recovered in September 2026.

	Sample	5th percentile	Median	95th percentile
Annualised return	22.6 %	5.8 %	22.7 %	39.5 %
Annualised volatility	24.7 %	21.9 %	24.6 %	28.2 %
Sharpe ratio	0.92	0.25	0.92	1.52
Maximum drawdown	−38.4 %	−60.8 %	−34.0 %	−21.9 %
Worst month	−16.9 %	−20.1 %	−14.4 %	−9.5 %
Positive months	61 %	51 %	61 %	70 %

Table 1: Net performance, full sample, January 2021 to September 2026, after costs and funding. Percentiles of a stationary block bootstrap of daily returns (blocks of 20 days, 2 000 paths). Returns in fraction of capital, not compounded.

	Sample	5th percentile	Median	95th percentile
Annualised return	9.0 %	−20.2 %	8.8 %	36.2 %
Annualised volatility	23.8 %	21.5 %	23.7 %	26.1 %
Sharpe ratio	0.38	−0.88	0.37	1.51
Maximum drawdown	−29.6 %	−58.8 %	−29.6 %	−16.4 %
Worst month	−16.9 %	−20.1 %	−13.5 %	−7.3 %
Positive months	67 %	38 %	57 %	76 %

Table 2: Net performance over the test alone, January 2025 to September 2026. Same method as Table 1.

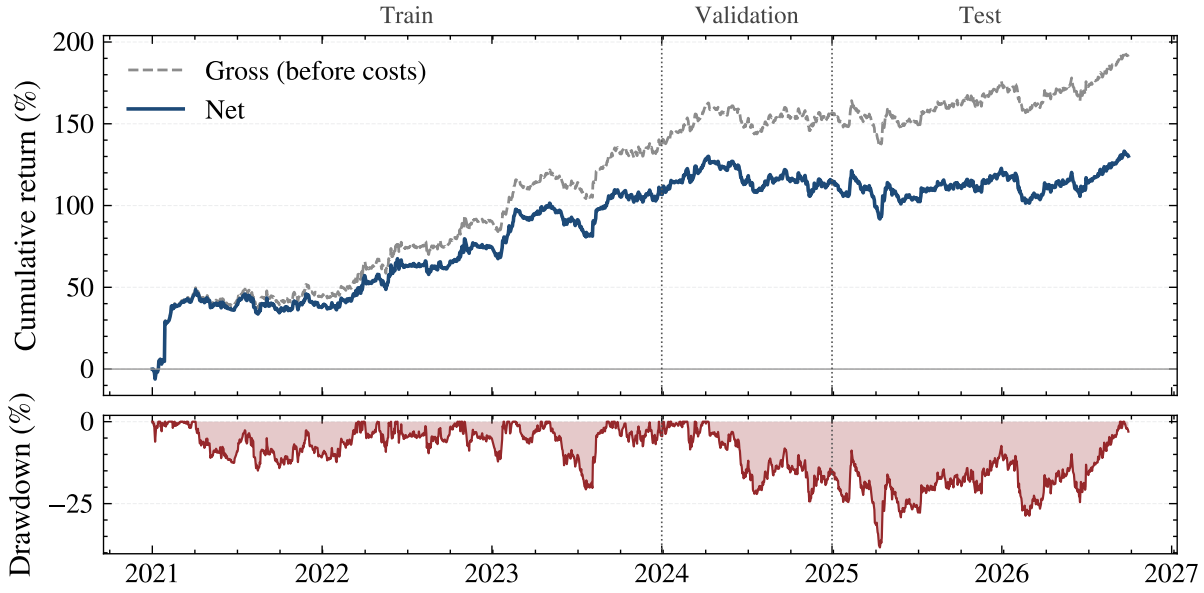


Figure 1: Cumulative net and gross return (top) and drawdown of the net return (bottom), in percent of capital, linear scale. Dotted lines mark the start of the validation and of the test.

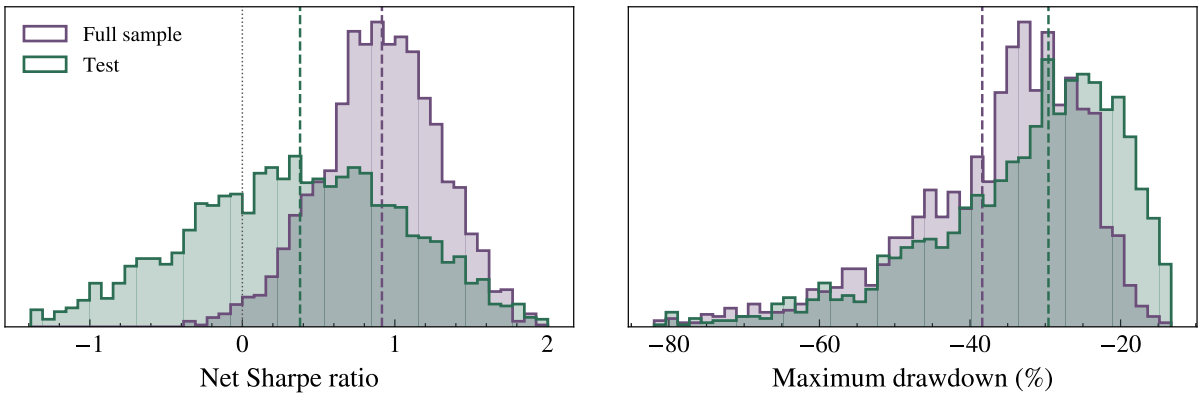


Figure 2: Bootstrap distributions of the net Sharpe ratio (left) and of the maximum drawdown (right). Dashed lines: the values measured on the sample.

3 Market exposure

The strategy is not a bet on the crypto market: its beta to BTC is 0.02 over the full sample and −0.02 over the test, and its correlation to BTC stays under 0.05 in absolute value. The alpha over BTC is significant on the full sample (21.9 % a year, t 2.16), carried by the train, and not on the test (9.1 %, t 0.47).

	Full sample	Test
Beta to BTC (daily)	0.02 (t 1.19)	−0.02 (t −0.74)
Correlation to BTC	0.05	−0.04
Alpha over BTC, annualised	21.9 % (t 2.16)	9.1 % (t 0.47)

Table 3: Exposure to the market. Regression of daily net returns on BTC daily returns; t-statistics with Newey-West standard errors (10 lags).

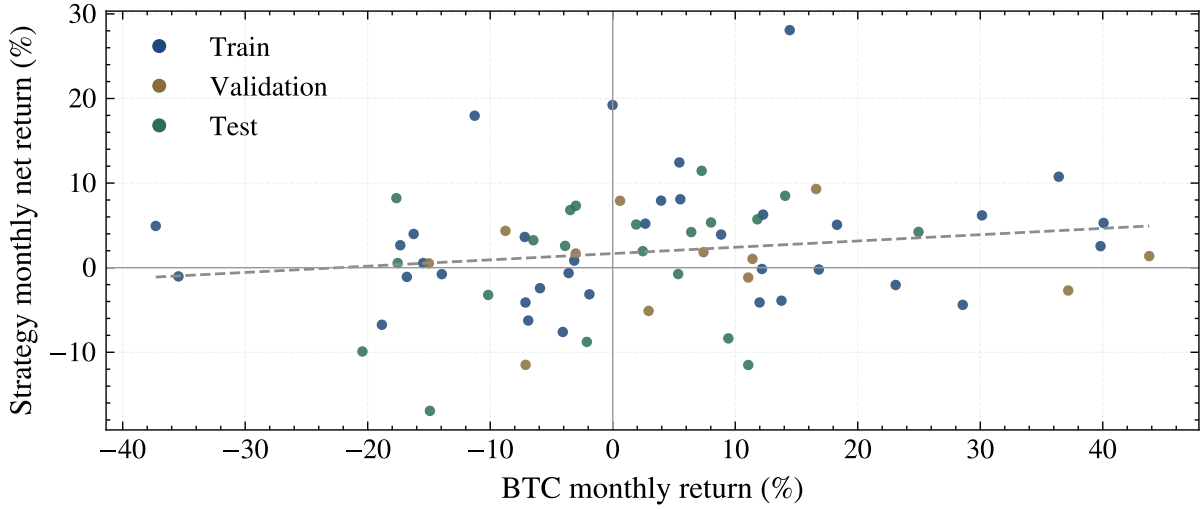


Figure 3: Monthly net return of the strategy against the monthly return of BTC. Dashed line: least-squares fit.

4 Where the profit comes from

The hypothesis says the long leg earns the money. It did on the train: the thin losers made 228 % of capital in price while the short leg lost 89 %. In the test the legs swapped: the long leg lost 45 % and the short leg, liquid losers that kept falling, made 97 %. Funding cost 17 % over the test against 4 % over the three years of train. The payer named in the hypothesis did not pay.

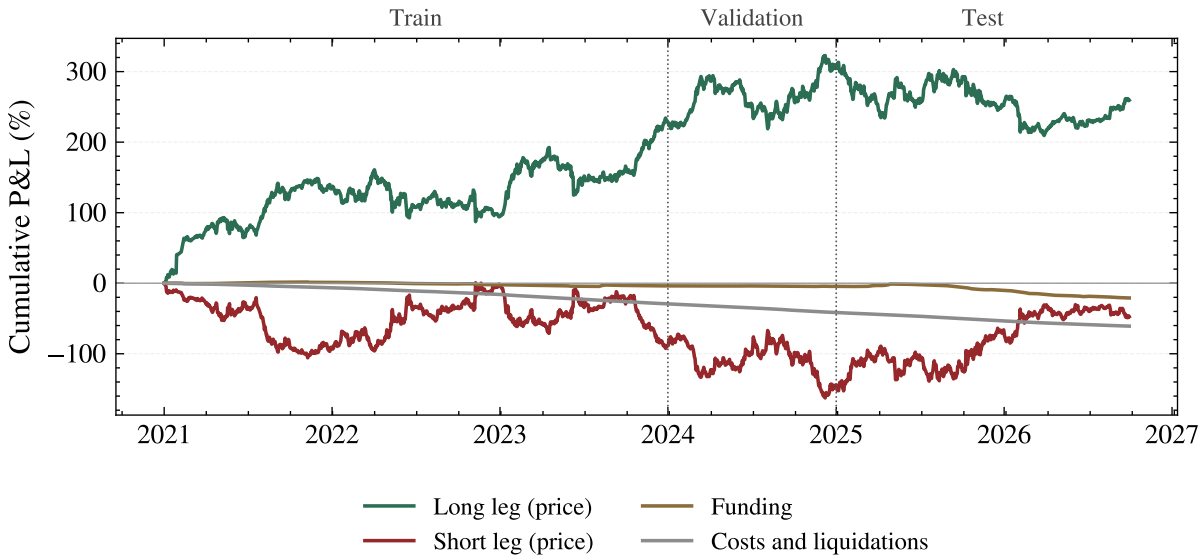


Figure 4: Cumulative P&L by source, in percent of capital: price of the long leg, price of the short leg, funding, and costs.

5 Concentration

The best 21 days of the full sample, 1 % of the days, make 97 % of its net P&L: without them the Sharpe ratio falls from 0.92 to 0.04. In the test, the best 7 days make more than twice the net P&L, and the best five names more than all of it. The same cut on BTC takes its Sharpe from 0.61 to −0.11: thick right tails are the nature of the market, but the result rests on rare events. The big days are regular, though: each month its two best days carry it, and the slowdown after 2023 came from the other days losing twice as much, not from fewer big days.

	Full sample	Test
Net Sharpe ratio, all days	0.92	0.38
Share of net P&L from the best 1 % of days	97 % (21 days)	235 % (7 days)
Net Sharpe ratio without the best 1 % of days	0.04	−0.58
Annualised return without the best 1 % of days	0.8 %	−12.4 %
Net Sharpe ratio without the best 5 % of days	−2.03	−2.69
BTC Sharpe ratio, all days / without its best 1 %	0.61 / −0.11	

Table 4: Concentration of the gains in time. A day removed is set to zero, as if the strategy had been flat that day. The BTC line gives the same cut on the market over the same span.

	Train	Validation	Test
Names traded	195	272	592
Share of net P&L, best name	16 %	20 %	42 %
Share of net P&L, best 5 names	41 %	77 %	164 %
Share of net P&L, best 10 names	64 %	135 %	267 %
Net P&L without the best 5 names, % of capital	80 %	5 %	−10 %

Table 5: Concentration of the gains across names: net P&L by name, best names first. A share above 100 % means the other names lost money together.

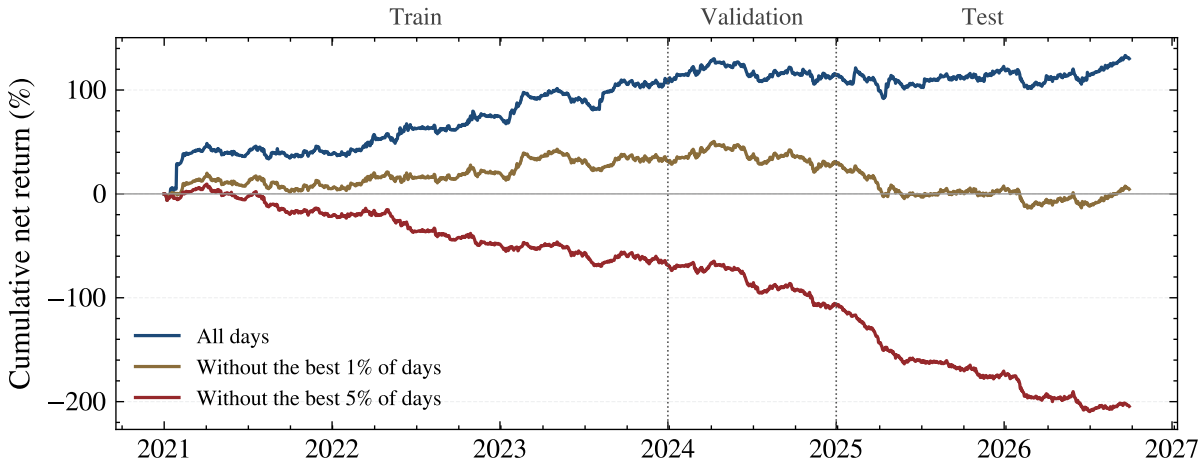


Figure 5: Cumulative net return with all days, and with the best 1 % and 5 % of days set to zero.

6 Implementation

The book holds about 30 names on each side in the train and about 95 in the test, at a gross exposure of 1.5 to 1.8, and keeps a position about four weeks. Costs take 10 to 12 % of capital a year. The break-even cost fell from 57 bp in the train to 20 bp in the validation and 23 bp in the test, against the 20 bp assumed on the thin leg.

	Train	Validation	Test
Names held long / short, average	34 / 31	59 / 56	95 / 92
Gross exposure, average	1.45	1.81	1.68
Holding period of a position, days	27	25	27
Turnover, times the capital a year	79.6	98.1	89.0
Cost drag, % of capital a year	−9.6 %	−11.8 %	−11.1 %
Break-even cost, bp one way	57.0	20.0	22.6
Days with a liquidation	18	21	0

Table 6: Implementation. The break-even cost is the one-way cost per unit traded at which the gross P&L of the period would be fully spent.

7 Robustness

The result holds a one-day delay and the choice of price, but not higher costs: with costs doubled or trades three days late, the validation Sharpe turns negative. Several sorting choices tried in the validation (independent sort, shorter beta window, equal-weight hedge, thinner cut) also turn it negative. The edge is thin compared with what it costs to trade.

	Train	Validation	Δ train	Δ validation
Frozen version, as tested	1.38	0.33		
Trades 1 day later	1.28	0.32	−0.10	−0.01
Costs × 2	1.01	−0.19	−0.37	−0.52
Last price instead of mark price	1.37	0.39	−0.01	+0.05
Trades 3 days later	1.14	−0.14	−0.24	−0.47

Table 7: Stress tests of the frozen version, run by the Red team before the test: net Sharpe ratio, and its change from the frozen version on the same period.

	Kind	Run on	Net Sharpe	Frozen version
winners-diagnostic	diagnostic	Train	−0.15	1.38
base-tiered	variant	Train	1.38	1.38
winners-diagnostic-tiered	diagnostic	Train	−0.15	1.38
depth-diagnostic-tiered	diagnostic	Train	−0.66	1.38
v1	variant	Validation	0.26	0.33
v2	variant	Validation	0.37	0.33
base-sort_ind	variant	Validation	−0.06	0.33
base-beta30	variant	Validation	−0.02	0.33
base-beta120	variant	Validation	0.64	0.33
base-lags10	variant	Validation	0.38	0.33
base-hedge_ew20	variant	Validation	−0.11	0.33
base-thin40	variant	Validation	−0.07	0.33
base-minleg5	variant	Validation	0.33	0.33

Table 8: Other versions run before the test, each on the period where the desk ran it: variants considered at the validation, controls and diagnostics that check the mechanism. They are not the strategy; the last column gives the frozen version on the same period for comparison.

8 Luck and multiple testing

Three versions of this strategy were run, and seven reversal strategies on crypto in all. On three years of train, the best of 5.3 independent tries with no skill would reach a Sharpe ratio of 0.71. The train net Sharpe of 1.38 clears that line by 0.67, and the probability that its true value exceeds it is 91 %. The line applies to the train, where the choices were made.

	Value
Tries counted for this strategy	3
Tries in the family, reversal on crypto (effective number)	7 (5.3)
Luck line: best Sharpe of 5.3 tries with no skill, 3.0 years	0.71
Train net Sharpe ratio / minus the luck line	1.38 / 0.67
Probability that the true train Sharpe exceeds the luck line	91 %
Whole desk: tries (effective number) / luck line	38 (9.6) / 0.76

Table 9: Multiple testing. Luck line: expected best annual Sharpe ratio of N_{eff} independent tries with no skill (Bailey and López de Prado, 2014); N_{eff} discounts tries by their correlation before costs. The probability is the deflated Sharpe ratio, corrected for the skewness (5.4) and the kurtosis (92) of daily train returns.

9 Risks and limits

The test book is not the train book: the universe tripled, and the strategy held about 95 names long on average against 34. The test run charged a flat 12.5 bp instead of the tiers. The daily returns of the train are very skewed (skewness 5.4), so the full-sample figures depend on a handful of days in 2021.

10 Decision

Kept in the library, on paper only. The full-sample t of 2.19 is carried by the train, and the test does not show the mechanism at work. To revisit after twelve months of paper trading: if the long leg keeps losing, the hypothesis is dead and what remains is a short on liquid losers, to be judged as such.

A Monthly returns

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2021	28.1	10.8	6.2	−3.2	−1.0	−2.4	5.1	−3.9	−4.1	5.3	3.6	−6.7	**37.6**
2022	−1.1	6.3	12.4	2.7	0.5	4.9	−0.2	−0.8	0.8	8.1	4.0	−0.6	**37.1**
2023	2.5	19.2	−2.0	5.2	−6.3	−4.1	−7.6	18.0	7.9	−4.4	3.9	−0.2	**32.2**
2024	7.9	1.4	9.3	0.5	1.0	−11.5	−5.1	4.4	1.8	−1.2	−2.7	1.7	**7.5**
2025	−8.4	8.2	−8.8	8.5	−11.5	2.0	5.3	3.2	−0.8	2.6	0.5	7.3	**8.3**
2026	−3.2	−16.9	5.1	5.7	6.8	−9.9	11.4	4.2	4.2				**7.5**

Table 10: Monthly net return, % of capital.

Runs: 20261024-083824-train-base, 20261023-164211-validation-base, 20261024-083811-test-base. Strategy file: crypto_thin_losers.py. Frozen version: base.